combat support, while in civilian sectors, they are transforming industries such as infrastructure inspection, agriculture, logistics, and disaster management through precision automation.

This chapter explores the technological foundation, operational architecture, and evolving applications of fourth-generation drones, emphasizing their role in autonomous decision-making, cooperative engagement, and digital integration. It also examines key global programs such as the US Skyborg, Europe's Future Combat Air System (FCAS), India's Ghatak, and China's Dark Sword, which exemplify the global race toward next-generation unmanned dominance. Furthermore, the chapter discusses the challenges of AI ethics, cybersecurity, regulatory adaptation, and human–machine collaboration, which continue to shape the responsible evolution of this powerful technology.

5.1.1 Definition

Fourth-generation drones are advanced unmanned aerial systems (UAS) that integrate artificial intelligence (AI), machine learning (ML), real-time data processing, and autonomous decision-making to perform complex missions with minimal or no human intervention. Unlike earlier generations that relied heavily on remote piloting or pre-programed commands, these drones possess the capability to perceive, analyze, and act independently based on situational data and mission objectives.

They are characterized by self-learning algorithms, swarm intelligence, sensor fusion, and networked interoperability, enabling coordinated operations among multiple unmanned and manned platforms across air, land, sea, and space. Equipped with AI-based navigation, adaptive mission planning, and autonomous threat response, fourth-generation drones can execute multi-domain operations from surveillance and electronic warfare to precision strikes and logistics.